

Bayesian Conditional-Quantile Forecasting with Fixed Deep Echo State Network Features

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Abstract

Conditional-quantile forecasting for nonlinear time series requires a representation that can retain long and multiscale temporal structure without making posterior computation prohibitively large. We condition on a fixed deep echo state network (DESN) feature map and place Bayesian quantile readouts on that representation. For a target probability p , the readout uses either the asymmetric Laplace (AL) working likelihood or its quantile-fixed generalized asymmetric Laplace (qf-GAL) extension, together with ridge or regularized-horseshoe shrinkage. Posterior computation uses partially Gibbs, slice, or Metropolis updates, while the available variational routines are described according to their actual implementation rather than as a single generic approximation. Multi-quantile forecasts are formed primarily from independently fitted single-level models and are made noncrossing as a reporting action on a declared probability grid; a joint composite extension is secondary. Case-specific simulation envelopes show gains in some distribution–quantile combinations but no uniform advantage for the flexible likelihood or any readout. A focused one-origin GloFAS study improves check loss, interval score, and a finite-grid CRPS approximation relative to the raw ensemble, although its 90% interval coverage is only 0.357 and its retrospective covariate contract is not a strict issue-time evaluation. The resulting claims are conditional on the fixed reservoir, the working likelihood, and the stated validation protocols.

1 Introduction

Forecast users often need conditional tails, central quantiles, and intervals rather than a conditional mean alone. The target at probability $p \in (0, 1)$ is the conditional quantile

$$q_{p,t+h} = \inf \{y : \Pr(Y_{t+h} \leq y \mid \mathcal{F}_t) \geq p\}, \quad (1.1)$$

where h is a forecast horizon and $\mathcal{F}_t$ is the information set available at the forecast origin. Quantile regression estimates this target through the check loss (Koenker and Bassett, 1978; Koenker, 2005). Time-series applications add a representation problem: the useful predictors may include nonlinear interactions, persistent dynamics, and information at several temporal scales.

Reservoir computing separates representation from readout. A recurrent system with fixed, randomly generated weights maps the observed history into a high-dimensional state, after which only a comparatively small output layer is estimated (Jaeger, 2001; Lukoševičius and Jaeger, 2009). Deep echo state networks stack such state maps and can expose dynamics on different effective time scales (Gallicchio et al., 2018). Freezing that map makes the Bayesian object explicit: uncertainty is quantified for a conditional-quantile readout given one realized feature representation, not for all possible reservoir weights.

We combine that fixed representation with AL and quantile-fixed generalized asymmetric Laplace working likelihoods and with ridge or regularized-horseshoe shrinkage. We use *qf-GAL* as the scientific name for the flexible likelihood. The legacy code and generated-table alias *exAL* is used only when reproducing those repository identifiers. Both likelihoods target a selected conditional quantile through their location parameter; neither converts model-based credible intervals automatically into frequentist confidence intervals under misspecification.

The paper makes four contributions. First, it states the forecast target and the fixed-DESN conditioning contract before introducing the likelihood. Second, it gives single-level Bayesian readouts whose high-dimensional coefficients are regularized while the feature extractor remains fixed. Third, it distinguishes the MCMC and variational algorithms actually used by the implementations, including their monitored objectives and zero-shape treatment. Fourth, it separates independent multi-level fitting, the primary strategy, from the later monotone reporting action and from an optional joint composite construction. Simulations and a focused GloFAS study are used to delimit where the approach helped and where calibration remained weak.

Section 2 defines the target and feature map. Section 3 presents the Bayesian readouts; Sections 4 and 5 describe computation and multi-quantile forecasting. Sections 6 and 7 report the simulation and GloFAS evidence, and Section 8 discusses scope and limitations.

2 Forecast Target and Fixed DESN Representation

Let $Y_t \in \mathbb{R}$ denote the response and let $\mathbf{u}_t$ collect predictors observed under a specified forecast-information contract. The generalized inverse in (1.1) permits discrete or mixed conditional distributions. When the distribution is continuous and strictly increasing near its p th quantile, it reduces to the familiar equation $\Pr(Y_{t+h} \leq q_{p,t+h} \mid \mathcal{F}_t) = p$. The probability level p is fixed when a single-level model is fitted.

2.1 A conceptual deep reservoir map

For layer $\ell = 1, \dots, L$, write a leaky reservoir recursion as

$$\tilde{\mathbf{h}}_{t,1} = f(\mathbf{W}_1 \mathbf{h}_{t-1,1} + \mathbf{U}_1 \mathbf{u}_t), \quad \mathbf{h}_{t,1} = (1 - a_1) \mathbf{h}_{t-1,1} + a_1 \tilde{\mathbf{h}}_{t,1}, \quad (2.1)$$

$$\tilde{\mathbf{h}}_{t,\ell} = f(\mathbf{W}_\ell \mathbf{h}_{t-1,\ell} + \mathbf{U}_\ell \mathbf{r}_{t,\ell-1}), \quad \mathbf{h}_{t,\ell} = (1 - a_\ell) \mathbf{h}_{t-1,\ell} + a_\ell \tilde{\mathbf{h}}_{t,\ell}, \quad (2.2)$$

where f is applied componentwise, $a_\ell \in (0, 1]$ is a leak rate, and $\mathbf{r}_{t,\ell-1}$ may be a reduced representation of the preceding layer. Biases and deterministic seasonal terms can be included in the inputs. The forecasting design vector stacks an intercept, selected reservoir states, direct covariates, and any declared transformations:

$$\mathbf{x}_{t,h} = \mathcal{R}_h(\mathcal{F}_t; \Omega), \quad \Omega = \{\mathbf{W}_\ell, \mathbf{U}_\ell, a_\ell : \ell = 1, \dots, L\}. \quad (2.3)$$

The function $\mathcal{R}_h$ also records preprocessing, lag construction, washout, reducers, and the rule for advancing predictors to horizon h .

In the analyses below, Ω and all feature-construction choices are frozen before Bayesian readout fitting. Random reservoir weights are scaled to a declared spectral-radius target and accepted under implementation checks, but a numerical radius rule is not by itself a universal proof of the echo-state property for every nonlinear, leaky, driven system. Different accepted reservoirs can yield different forecasts. Thus every posterior statement is conditional on the realized map $\mathcal{R}_h$.

Inferential contract. Observed history and permitted predictors are transformed once into a fixed design $\mathbf{x}_{t,h}$. Bayesian inference then concerns the readout coefficients, likelihood scale and shape, and shrinkage parameters conditional on that design. Reservoir-selection variability and data-dependent feature-selection variability are outside the reported posterior unless explicitly added by a separate validation layer.

This separation is computationally useful: the nonlinear recurrent representation is generated without back-propagating through a large Bayesian recurrent network. It is also a limitation. The posterior can be sharp even when reservoir selection, the forecast-covariate contract, or the feature map is misspecified.

3 Bayesian Conditional-Quantile Readouts

For one probability level p and horizon h , the conditional-quantile readout is

$$\mu_{p,t,h} = \alpha_{p,h} + \mathbf{x}_{t,h}^\top \boldsymbol{\beta}_{p,h}, \quad Q_p(Y_{t+h} \mid \mathcal{F}_t, \Omega) = \mu_{p,t,h}. \quad (3.1)$$

Separate horizon-specific coefficients may be fitted, or a common readout may be evaluated on horizon-specific features. We suppress h below when it is not needed.

3.1 AL working likelihood

Let $\rho_p(r) = r\{p - \mathbf{1}(r < 0)\}$ be the check loss. The AL working density with location μ and scale $\sigma > 0$ is

$$f_{\text{AL}}(y \mid \mu, \sigma, p) = \frac{p(1-p)}{\sigma} \exp \left[-\rho_p \left(\frac{y - \mu}{\sigma} \right) \right]. \quad (3.2)$$

Its location is the p th quantile. A normal–exponential augmentation can be written

$$\begin{aligned} Y_t \mid v_t, \Theta_p &\sim \mathcal{N}(\mu_{p,t} + A_p v_t, B_p \sigma v_t), & v_t \mid \sigma &\sim \text{Exp}(1/\sigma), \\ A_p &= \frac{1 - 2p}{p(1 - p)}, & B_p &= \frac{2}{p(1 - p)}. \end{aligned} \quad (3.3)$$

This representation supports conditionally Gaussian coefficient updates. The likelihood scale controls the spread of the working observation model; it is not itself “uncertainty in the quantile.”

3.2 qf-GAL working likelihood

The qf-GAL extension (Yan et al., 2025) adds a bounded shape parameter γ while keeping the location at the selected quantile. One useful augmentation is

$$s_t \sim \mathcal{N}^+(0, 1), \quad v_t \mid \sigma \sim \text{Exp}(1/\sigma), \quad (3.4)$$

$$Y_t \mid s_t, v_t, \Theta_p \sim \mathcal{N}(\mu_{p,t} + \lambda(\gamma)\sigma s_t + A\{p, \gamma\}v_t, B\{p, \gamma\}\sigma v_t). \quad (3.5)$$

With Φ the standard normal distribution function, define

$$g(\gamma) = 2\Phi(-|\gamma|) \exp(\gamma^2/2), \quad p_\gamma = \mathbf{1}(\gamma < 0) + \frac{p - \mathbf{1}(\gamma < 0)}{g(\gamma)}, \quad (3.6)$$

$$C(\gamma) = \{\mathbf{1}(\gamma > 0) - p_\gamma\}^{-1}, \quad \lambda(\gamma) = C(\gamma)|\gamma|, \quad (3.7)$$

$$A\{p, \gamma\} = \frac{1 - 2p_\gamma}{p_\gamma(1 - p_\gamma)}, \quad B\{p, \gamma\} = \frac{2}{p_\gamma(1 - p_\gamma)}. \quad (3.8)$$

The admissible bounded support for γ keeps $p_\gamma \in (0, 1)$. The implementation assigns the AL constants at $\gamma = 0$ rather than evaluating the branch formula there. The resulting likelihood has the AL value and continuous limit at zero, and each nonzero-sign branch is smooth in its interior. For $p \neq 1/2$, however, the one-sided derivatives of p_γ at zero differ; even at the median a global twice-differentiability claim is not justified. Algorithms therefore treat the two nonzero branches explicitly or handle AL as a separate model. We do not rely on a globally regular Hessian across $\gamma = 0$.

Both (3.2) and (3.5) are proper working likelihoods under their stated parameterizations. Their quantile interpretation is the main inferential reason to use them. If the conditional response law is not AL or qf-GAL, posterior credible and predictive bands remain model-based and need not have nominal repeated-sampling coverage; sandwich or other calibrated corrections are separate procedures (Sriram et al., 2013; Sriram and Ramamoorthi, 2017; Yang et al., 2016; Ji et al., 2025). The shape parameter offers additional residual asymmetry and tail behavior, but the empirical studies below do not support a uniform gain over AL.

3.3 Shrinkage priors

The fixed reservoir may produce a design whose dimension is large relative to the effective sample size. A ridge readout uses

$$\beta_p \mid \kappa_p^2 \sim \mathcal{N}(\mathbf{0}_d, \kappa_p^2 \mathbf{I}_d), \quad \kappa_p^2 \sim \text{IG}(a_\kappa, b_\kappa), \quad (3.9)$$

with a separately regularized intercept. The regularized horseshoe (RHS) instead uses

$$\begin{aligned} \beta_{p,j} \mid \tau_p, \lambda_{p,j}, c_p &\sim \mathcal{N}(0, V_{p,j}), & V_{p,j} &= \tau_p^2 \frac{c_p^2 \lambda_{p,j}^2}{c_p^2 + \tau_p^2 \lambda_{p,j}^2}, \\ \lambda_{p,j} &\sim \text{C}^+(0, 1), & \tau_p &\sim \text{C}^+(0, \tau_{0,p}), \end{aligned} \quad (3.10)$$

with a prior on the finite slab scale c_p . The global scale shrinks the whole readout, local scales protect selected features, and the slab prevents unbounded escape. These priors regularize the readout only; they do not average over reservoir maps. The construction follows the horseshoe and regularized-horseshoe distinction in Carvalho et al. (2010); Piironen and Vehtari (2017); the supplement gives the auxiliary-variable form used by the sampler.

4 Posterior Computation and Approximation

For fixed p , fixed design X , and readout parameters Θ_p , the posterior is

$$\pi(\Theta_p, \mathbf{v}, \mathbf{s} \mid \mathbf{y}, X, p) \propto \prod_{t=1}^T p(y_t \mid v_t, s_t, \Theta_p, X, p) p(\mathbf{v}, \mathbf{s} \mid \Theta_p) p(\Theta_p), \quad (4.1)$$

where $\mathbf{s}$ is omitted for AL. Computation exploits the augmented conditionals but is not uniformly conjugate.

4.1 MCMC

Conditionally Gaussian and inverse-gamma blocks are sampled directly when those full conditionals are available. Generalized inverse Gaussian (GIG) full conditionals are realized in the production code by invariant slice transitions on the log scale rather than by an independent GIG random-number generator. Nonconjugate shape blocks use slice or Metropolis updates. In the qf-GAL joint-validation kernel used for the reported MCMC evidence, the scale is integrated out of the shape target, the bounded shape is updated by slice sampling on a support-logit coordinate, and the scale is then updated from its conditional GIG target with the log-scale slice transition. The overall sampler is therefore a partially Gibbs, slice-within-Gibbs algorithm. Calling it “exact MCMC” would mean only that its stated transition kernel leaves the specified augmented working posterior invariant; it would not make a finite chain exact, remove Monte Carlo error, or validate the working likelihood as the data-generating law.

We assess chains using multiple diagnostics rather than acceptance or completion alone: split- $\hat{R}$ and effective sample size for scalar blocks, rank or trace checks, finite-value and support checks, quantile-path stability, and repeated-chain confirmation when required. A numerical **PASS** is evidence about the sampled target and chain behavior under the specified thresholds, not a scientific guarantee of forecast improvement.

4.2 Implementation-specific variational routines

The available variational algorithms do not share one mathematical objective, so we avoid using “VB–Laplace–Delta” as a blanket label.

- The Article-v2 point-shape routine optimizes a point value of γ and updates remaining blocks with a coordinate approximation. Its reported monitor is a partial coordinate expected-log-joint quantity, not a complete evidence lower bound (ELBO), and it does not use a joint Hessian Laplace factor for the shape and scale.
- The Article-v2 branch-structured routines integrate over explicit sign branches for the shape and conditionally update scale terms. Their monitors are likewise partial coordinate quantities; omitted local or point-mass entropy terms preclude interpreting them as complete ELBOs.
- The external single-level engine used in some validation paths does construct a joint numerical-Hessian Gaussian approximation on transformed shape and log-scale coordinates and applies deterministic second-order Delta calculations in the spirit of nonconjugate variational inference (Wang and Blei, 2013). Curvature repair and eigenvalue clipping can be invoked, so this remains an implementation-specific Laplace–Delta approximation rather than a general property of the Article-v2 routines.
- The GloFAS analysis below uses a separate AL-only variational readout. It reports a first-order-Delta expected-log-joint monitor, not a complete ELBO, and contains no qf-GAL shape approximation.

These distinctions matter most near $\gamma = 0$, where the qf-GAL parameterization is branchwise rather than globally twice differentiable. A Hessian computed within a branch can be useful, but it should not be presented as curvature of one globally regular objective across zero. For substantive comparisons in this paper, the MCMC results provide the main reference where available; variational results are interpreted as algorithm-specific approximations and are checked against deterministic fit, support, and forecast contracts.

5 From Single-Level Inference to Multi-Quantile Forecasts

Let $\mathcal{P} = \{p_1 < \dots < p_K\}$ be a declared grid. The primary construction fits the model in Sections 3–4 separately at each p_k . For posterior draw m ,

$$q_{p_k, t+h}^{(m)} = \alpha_{p_k, h}^{(m)} + \mathbf{x}_{t, h}^\top \boldsymbol{\beta}_{p_k, h}^{(m)} \quad (5.1)$$

is a conditional-quantile forecast given the fixed feature path. Independent fits are parallel and preserve the meaning of each single-level working likelihood, but their raw forecasts can cross.

5.1 Noncrossing as a reporting action

For each origin, horizon, and posterior summary or draw, we project the vector of raw grid values $\tilde{\mathbf{q}}$ onto the monotone cone:

$$\hat{\mathbf{q}} = \arg \min_{q_1 \leq \dots \leq q_K} \sum_{k=1}^K w_k (q_k - \tilde{q}_k)^2, \quad w_k > 0. \quad (5.2)$$

Weighted isotonic regression computes (5.2) efficiently (Barlow and Brunk, 1972; Chernozhukov et al., 2010). This is a deterministic reporting and decision rule. It changes the displayed quantile vector; it does not retroactively define a joint posterior over levels or prove that the independently fitted likelihoods were coherent. We therefore retain raw crossing counts as diagnostics and report whether scoring uses raw or projected forecasts.

Between declared levels, quantiles are linearly interpolated unless stated otherwise. We do not infer unrestricted tail behavior beyond $[p_1, p_K]$. A score computed by integrating check loss over this finite grid is reported as a *finite-grid CRPS approximation*, not as the exact continuous-distribution CRPS:

$$\widehat{\text{CRPS}}_{\mathcal{P}}(F, y) = 2 \sum_{k=1}^K \omega_k \rho_{p_k} \{y - \hat{q}_{p_k}\}, \quad (5.3)$$

where ω_k records the declared quadrature rule. Lower values indicate better forecasts, as they do for check loss and interval score.

5.2 One-step and recursive forecast propagation

For a one-step forecast, every lagged response in $\mathbf{x}_{t,1}$ is observed at the forecast origin. At longer horizons, any unavailable response lag must be supplied by a declared recursive rule. In the multi-quantile path construction used here, all raw ordinates at a horizon are evaluated against one common lag history, projected by (5.2), and interpolated on $\mathcal{P}$. Drawing one $U \sim \text{Unif}(0, 1)$ and setting $Y^* = Q^*(U)$ then supplies the same synthesized response to every level-specific state recursion for the next horizon. This common-history rule avoids assembling a quantile curve from incompatible future response paths. It is an action-induced draw from the reported quantile function, not a draw obtained by normalizing or mixing the level-specific working likelihoods. Future

exogenous predictors must come from the issue-time information set or a declared scenario; use of realized future predictors defines a retrospective replay.

5.3 A secondary joint composite extension

A joint readout can share a baseline and represent level-specific departures, for example

$$\beta_{p_k} = \beta_0 + \delta_k, \quad \delta_k \mid \omega_k^2 \sim \mathcal{N}(\mathbf{0}_d, \omega_k^2 \mathbf{I}_d), \quad (5.4)$$

with shrinkage linking neighboring levels (Kohns and Szendrei, 2025). The implemented joint target multiplies level-specific working-likelihood contributions for the same response across p_k . Without an explicit tempering or calibration construction, this is an untempered composite working posterior: it is not the posterior induced by a unique normalized response density. Soft coherence penalties and the final monotone projection can reduce crossings, but they do not change that interpretation. We therefore treat the joint model as a secondary empirical extension rather than the organizing model of the paper.

The forecast protocol must be fixed independently of the readout: training endpoint, origins, horizons, refitting frequency, predictor availability, preprocessing, and any hyperparameter selection all define the estimand. Reported application comparisons use check loss, central interval score, the finite-grid approximation in (5.3), crossing counts, and interval coverage. Coverage is reported with its nominal target, because improved relative score can coexist with substantial undercoverage.

6 Simulation Evidence

The simulations ask three limited questions: whether fixed nonlinear features can improve selected conditional-quantile forecasts, whether the qf-GAL shape extension helps consistently relative to AL, and whether a joint composite readout reliably improves the independent fits. They are not a proof of universal dominance.

6.1 Single-level comparisons

The main single-level study uses Gaussian, Laplace, and Gaussian-mixture innovation families and target probabilities $p \in \{0.05, 0.25, 0.50\}$. Baseline dynamic quantile linear models are compared with AL-RHS and qf-GAL-RHS readouts on fixed DESN features. The shared protocol uses train-only preprocessing, and forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000, leads 1–30, and origin stride 30.

Tables 1–3 are deliberately metric-level evidence displays. Each Q-DESN row is a fixed, case-specific envelope: entries in one displayed row can come from different calibrated specifications or seeds, and a confirmed criterion can use a predeclared repeated-chain aggregate. The rows therefore must not be read as the joint performance vector of one coherent fitted model. The design uses one dynamic response realization per family-level cell rather than a Monte Carlo

ensemble of independently generated series, so the tables describe these cases and do not estimate a repeated-simulation sampling distribution. Dagger symbols retain contributing warning or failure diagnostics.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	3.036	3.829	1.103	3.811	2.209	3.328	2.590	1.161	4.029
exDQLM	1.935	16.23	4.313	2.344	6.634	4.132	2.808	1.637	4.109
Q-DESN AL-RHS [†]	2.779	8.410	1.264	2.176	2.191	3.292	1.523	2.104	4.073
Q-DESN exAL-RHS [‡]	2.149	2.863	1.080	1.366	2.415	3.317	1.128	1.971	4.059

Table 1: MCMC single-quantile fit-and-forecast comparison for the Gaussian simulation family. The Q-DESN entries form a fixed case-specific metric envelope and can therefore draw different criteria from different calibrated fits. A confirmed criterion may instead use its predeclared repeated-chain aggregate, as identified in the metric-level record. Q-DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger or double dagger records a contributing WARN or FAIL diagnostic, respectively.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	3.663	9.791	1.968	2.850	3.520	4.548	1.774	1.337	5.082
exDQLM	5.677	22.44	5.154	1.710	6.811	5.106	1.774	2.013	5.206
Q-DESN AL-RHS [†]	5.420	3.893	1.873	2.256	1.434	4.379	1.353	0.7553	5.008
Q-DESN exAL-RHS [‡]	6.409	2.362	1.855	1.383	0.6696	4.343	1.075	0.6772	5.008

Table 2: MCMC single-quantile fit-and-forecast comparison for the Laplace simulation family. The Q-DESN entries form a fixed case-specific metric envelope and can therefore draw different criteria from different calibrated fits. A confirmed criterion may instead use its predeclared repeated-chain aggregate, as identified in the metric-level record. Q-DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger or double dagger records a contributing WARN or FAIL diagnostic, respectively.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	2.623	3.003	1.505	4.740	2.487	4.540	2.226	1.932	5.556
exDQLM	2.554	23.91	5.857	1.381	8.941	5.649	2.408	2.207	5.607
Q-DESN AL-RHS [†]	3.154	3.006	1.564	1.855	0.9122	4.467	1.271	2.368	5.585
Q-DESN exAL-RHS [‡]	3.354	2.913	1.527	1.380	3.396	4.586	0.9177	2.562	5.610

Table 3: MCMC single-quantile fit-and-forecast comparison for the Gaussian mixture simulation family. The Q-DESN entries form a fixed case-specific metric envelope and can therefore draw different criteria from different calibrated fits. A confirmed criterion may instead use its predeclared repeated-chain aggregate, as identified in the metric-level record. Q-DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger or double dagger records a contributing WARN or FAIL diagnostic, respectively.

The pattern is heterogeneous. The qf-GAL readout has the lowest displayed forecast check loss in

several Gaussian and Laplace cells, while AL or a baseline is best in others. In the Gaussian-mixture family, for example, AL has the lowest displayed check loss at $p = 0.25$, whereas the baseline is lowest at $p = 0.05$ and $p = 0.50$. The conclusion is therefore conditional improvement, not a uniform qf-GAL or Q-DESN advantage. A subsequent paired confirmation revisited the retained flexible-likelihood result and found no confirmed improvement, so it did not alter these article metrics.



Figure 1: Metric-level display for the case-specific single-quantile MCMC envelope. Color comparisons should be read within the declared family, probability, and criterion; they do not turn the envelope into a single fitted-model result.

6.2 Secondary joint comparison

The joint study spans eight mechanisms and compares independent and joint AL-RHS and qf-GAL-RHS readouts. Table 4 reports forecast MAE, fit-window MAE, and raw adjacent-level crossings. Eleven qf-GAL entries use newly qualified M0 results satisfying prespecified finite-path and quantile-stability criteria; five retain previously verified qf-GAL results because their replacement paths did not satisfy those criteria. The table is consequently a qualified scenario summary, not a homogeneous fresh rerun with common winner-runner-up Monte Carlo uncertainty.

The joint AL readout has the lowest forecast MAE for the Laplace, nonlinear-reservoir-friendly, and normal mechanisms, while independent AL is lowest for the asymmetric-Laplace-tail, regime-

Scenario	Mechanism	Joint QDESN AL-RHS	Independent QDESN AL-RHS	Joint exQDESN exAL-RHS	Independent exQDESN exAL-RHS
Asymmetric Laplace Tail	Asymmetric Laplace; seasonal AR(1) location-scale	0.099 (0.084; 1)	0.083 (0.088; 7)	0.093 (0.081; 0)	0.089 (0.077; 0)
Gaussian-Mixture Bridge	Gaussian mixture; seasonal AR(1) location-scale	0.096 (0.097; 0)	0.099 (0.101; 0)	0.106 (0.096; 0)	0.094 (0.095; 0)
Laplace Bridge	Laplace; seasonal AR(1) location-scale	0.085 (0.089; 0)	0.094 (0.091; 7)	0.111 (0.113; 0)	0.106 (0.117; 0)
Nonlinear Reservoir Friendly	Gaussian mixture; nonlinear reservoir dynamics	0.104 (0.109; 0)	0.116 (0.112; 1)	0.157 (0.151; 0)	0.149 (0.143; 0)
Normal Bridge	Gaussian; seasonal AR(1) location-scale	0.081 (0.082; 0)	0.082 (0.082; 2)	0.115 (0.097; 0)	0.104 (0.101; 0)
Persistent Heavy Tail	Student-t; persistent seasonal AR(1) dynamics	0.131 (0.120; 0)	0.124 (0.118; 0)	0.130 (0.103; 0)	0.106 (0.100; 0)
Regime Shift	Student-t; regime-shift location-scale	0.153 (0.085; 0)	0.096 (0.091; 6)	0.185 (0.090; 0)	0.113 (0.087; 0)
Student-t Location-Scale	Student-t; seasonal AR(1) location-scale	0.080 (0.062; 0)	0.071 (0.065; 1)	0.116 (0.093; 0)	0.099 (0.092; 0)

Table 4: Scenario-level MCMC confirmation for the joint multi-quantile validation study. Entries are forecast MAE with fit-window MAE and the number of raw forecast-window adjacent-level crossings in parentheses, written as forecast MAE (fit MAE; raw crossings). Boldface marks the lowest forecast MAE within each scenario. QDESN uses the AL working likelihood, exQDESN uses the exAL working likelihood, and RHS denotes the regularized horseshoe prior. All scores use the monotone quantile-grid contract; raw crossings are retained only as pre-contract diagnostics. Eleven qf-GAL entries use newly qualified M0 results satisfying prespecified finite-path and quantile-stability criteria; five use previously verified MCMC results because their replacement paths did not satisfy those criteria.

shift, and Student- t location-scale mechanisms. A qf-GAL readout is lowest in the Gaussian-mixture and persistent-heavy-tail mechanisms. Raw crossing counts are concentrated in the independent AL fits and fall after the common monotone reporting contract. These results support the joint model as a useful option in some scenarios, but not as a uniformly superior likelihood or as a coherent generative multi-quantile posterior.

7 GloFAS Reference Application

We use one GloFAS river-discharge example as a focused reference application rather than as a broad operational benchmark (Alfieri et al., 2013; Harrigan et al., 2023). The analysis concerns one gauge, one forecast origin (December 25, 2022), a 28-day horizon, and 51 raw ensemble members. Seven independently fitted AL readouts target $p \in \{0.05, 0.15, 0.35, 0.50, 0.65, 0.80, 0.95\}$. Separate fixed feature maps describe the reference path and the GloFAS-minus-reference discrepancy. For each level p ,

$$q_{p,t}^{\text{ref}} = \mathbf{x}_t^{\text{ref}} \top \boldsymbol{\beta}_p, \quad (7.1)$$

$$d_{p,t}^{\text{glo}} = \mathbf{x}_t^{\text{disc}} \top \boldsymbol{\alpha}_p, \quad q_{p,t}^{\text{glo}} = q_{p,t}^{\text{ref}} + d_{p,t}^{\text{glo}},$$

$$q_{p,t+h}^{\text{ref}} = q_{p,t+h}^{\text{glo}} - d_{p,t+h}^{\text{glo}}. \quad (7.2)$$

The issued ensemble informs $q_{p,t+h}^{\text{glo}}$, whereas the corrected target is the reference-process quantile $q_{p,t+h}^{\text{ref}}$. The response is transformed with $\log(1 + y)$ during fitting, and the scores reported here remain on that transformed scale.

The held-out future discharge responses are not used to construct forecast states or select the fitted readout. In that sense the protocol is response-blinded. The retrospective feature contract nevertheless permits realized-future precipitation and soil information to be blended with forecast covariates. It is therefore not a strict issue-time-covariate evaluation, and the result should not be described as prospective operational performance. The reservoir features are built in Article-v2 and the readout is the separate AL-only, first-order-Delta variational routine described in Section 4; the external engine’s qf-GAL Hessian approximation is not used for these reported fits.

Metric	Corrected Q-DESN	Raw GloFAS
Mean check loss	0.5654	0.7639
Mean central-interval score	10.8032	25.2028
Finite-grid CRPS approximation	1.1319	1.4424
Mean nominal-90% interval coverage	0.357	0.000

Table 5: One-origin, 28-day GloFAS comparison. Lower values are better for check loss, interval score, and the finite-grid CRPS approximation. Coverage is shown against a nominal 0.90 target; it is not an optimization score.

Mean check loss averages $28 \times 7 = 196$ horizon–level pairs; interval score and coverage average the 28 central intervals; and the CRPS approximation uses trapezoidal integration over the seven fitted levels. The correction lowers mean check loss from 0.7639 to 0.5654, mean interval score from 25.2028 to 10.8032, and the declared finite-grid CRPS approximation from 1.4424 to 1.1319. Those relative gains do not imply good absolute calibration: mean coverage of the displayed nominal-90% interval is 0.357, far below 0.90, although the raw ensemble’s corresponding value is 0.000. The result is a useful correction example with a serious undercoverage limitation, not a calibrated distributional-forecasting claim. The variational monitor converged in a median of 103 and a maximum of 356 iterations, but those counts concern an expected-log-joint stopping rule rather than a complete ELBO certificate (Gneiting and Raftery, 2007; Gneiting, 2011).

Transfer benchmark: PriceFM. As a separate transfer check, the same fixed-feature quantile-readout idea was compared with released PriceFM outputs (Yu et al., 2026). On the aligned 114-cell panel, Q-DESN has lower average quantile loss in 66 cells and is within the prespecified near-equivalence threshold in another 12; mean losses are 6.826 and 7.039 for Q-DESN and the released PriceFM checkpoint. The paper-reported PriceFM full-shot AQL of 5.80 uses a different fitted-model and aggregation pipeline and is contextual only, so it is not ranked against the aligned replay. Every replay policy uses realized-ex-post target load, solar, and wind leads; graph-feature rows additionally use realized-ex-post neighbor leads. Accordingly, the aligned results do not establish strict issue-time performance, an operational ranking, or evidence for the joint multi-quantile construction.

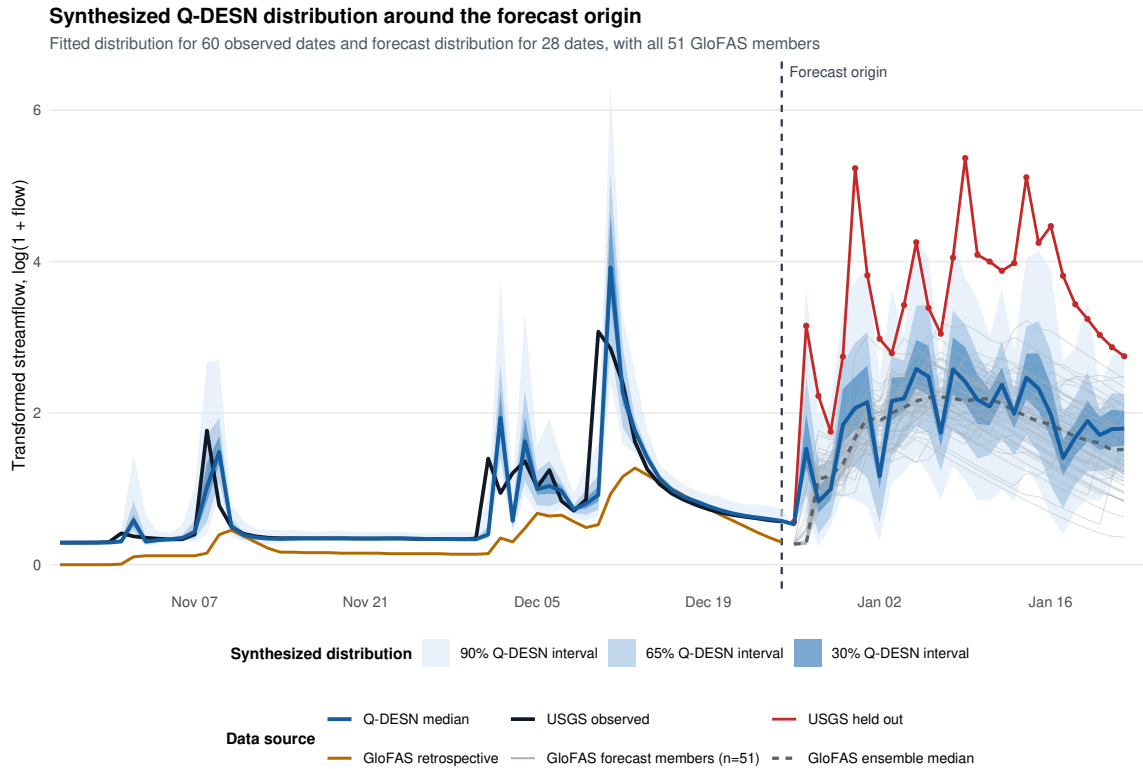


Figure 2: Observed discharge, raw GloFAS ensemble summaries, and corrected conditional-quantile forecasts over the reference window. The figure is response-blinded with respect to held-out discharge but uses the retrospective covariate contract described in the text.

8 Discussion

The central proposal is deliberately modular. A frozen deep reservoir compresses nonlinear temporal history; a Bayesian AL or qf-GAL readout targets one conditional quantile; ridge or RHS priors regularize the high-dimensional output layer; and independently fitted levels are reconciled only at the reporting stage. This organization keeps the inferential target visible and makes clear which uncertainty is and is not propagated.

The evidence supports a qualified conclusion. Fixed DESN features improve selected distribution-quantile cases in the simulations, but the best method changes with the innovation family, target level, and metric. The qf-GAL extension is useful in some cells and inferior in others; a separate repeated-chain check did not establish a new gain over the retained result. In the frozen eight-mechanism study, joint AL produced one raw adjacent-level crossing versus 24 for independent AL, both qf-GAL constructions produced none, and the common reporting action removed all reported crossings; forecast-MAE gains still depended on the mechanism. The GloFAS correction improves three relative scores at one gauge and origin, while its 0.357 coverage and retrospective realized-future covariate blend prevent a strong calibration or operational claim. The PriceFM transfer is also retrospective: every policy uses realized-ex-post target load, solar, and wind leads, and graph rows additionally use neighbor leads.

Several limitations follow directly. First, posterior uncertainty is conditional on a selected random reservoir and does not include reservoir-selection variability. Second, AL and qf-GAL are working likelihoods; model-based credible and predictive bands do not automatically achieve frequentist coverage under misspecification. Third, qf-GAL is branchwise at zero, so globally smooth-Hessian arguments are inappropriate. Fourth, Article-v2 and external variational routines optimize or monitor different quantities, some of which are partial expected-log-joint criteria rather than complete ELBOs. Fifth, the single-level simulation tables are heterogeneous metric envelopes based on one dynamic realization per cell, and the joint table retains some historical qualified results. Sixth, finite-grid CRPS values depend on the chosen probability grid, interpolation, quadrature, and tail convention. Finally, the application evidence is narrow and does not evaluate a fully issue-time operational pipeline.

These limitations suggest concrete next steps: repeat the simulation design over independently generated series and independently generated reservoirs; evaluate reservoir averaging or selection-aware uncertainty; compare working-likelihood credible bands with calibrated predictive procedures; study zero-shape qf-GAL inference with explicitly branch-aware approximations; temper or otherwise calibrate the joint composite target; and run multi-origin, multi-gauge experiments using archived issue-time covariates only. Such studies would test whether the conditional improvements reported here persist beyond the fixed-feature and retrospective contracts that define the present analysis. If that evidence becomes available, the joint composite construction and the multi-origin calibration application would each merit separate treatment rather than being enlarged into additional central claims here.

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